

Joseph Zou

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EDUCATION

Vanderbilt University

Nashville, TN

Bachelor of Science in Computer Science (GPA: 3.85/4.00)

Aug. 2021 – May 2025

- **Relevant Courses:** Database Management Systems, Operating Systems, Algorithms, Computer Networks, Artificial Intelligence, Machine Learning, Data Science, Security & Privacy in Pervasive Environments (PhD Level)

EXPERIENCE

Software Engineer Intern

June 2024 – April 2025

Kinetik Care

New York City

- Developed an internal admin portal using **Retool**, **JavaScript**, and **MongoDB**, integrating company services to centralize client onboarding and manage the full organizational dataset, reducing onboarding time by **16%**.
- Worked with internal APIs to map and analyze relationships between Transportation Providers, Brokers, and Health Plan partners, improving data visibility and operational efficiency.

Software Engineer

February 2024 – August 2024

Vanderbilt University Medical Center - Department of Biomedical Informatics

Nashville, TN

- Led end-to-end development of a full-stack **React Native** application, overseeing architecture, product design, UI/UX, and database integration through the **RedCap API**.
- Trained and deployed a **YOLOv7** object detection model on curated pill image datasets to **Docker**, automating blister pack pill counting.

Software Engineer

November 2023 – Present

Mother to Mother

Nashville, TN

- Developed a fully integrated **Progressive Web App** and Admin portal using Prisma ORM, deployed on AWS EC2 with RDS, replacing **10+ years** of fragmented spreadsheets and paper records, reducing data entry and retrieval time by **70%**.
- Delivered and maintained a mission-critical platform supporting **5,000+ children and families**, streamlining workflows and reporting to save **20+ staff hours weekly**—becoming indispensable to daily operations.

Undergraduate Teaching Assistant

Vanderbilt University

Nashville, TN

- **CS 2201: Program Design and Data Structures** — 167 students *Jan 2025 – Apr 2025*
- **CS 3265: Database Management Systems** — 63 students *Aug 2024 – Dec 2024*
- Led office hours, one-on-one Zoom sessions, exam proctoring, and project evaluations in **Python** and **SQL**.

PROJECTS

PassWin | *AWS, React, MongoDB*

- Built a zero-knowledge password manager Chrome extension with Argon2id key derivation, AES-GCM encryption, and custom session authentication.
- Deployed on AWS EC2 with Nginx, SSL/TLS, MongoDB Atlas, and GitHub Actions CI/CD, delivering secure, end-to-end encrypted storage.

DeskMate | *Python, Tkinter, Design Patterns*

- Built a desktop app with animated sprites delivering productivity tools (motivational quotes, Pomodoro timer, dictionary, weather) using design patterns, API integrations, and caching for smooth, scalable performance.

LeetHub | *OAuth, Redis, Docker*

- Developed a tool that automates uploading LeetCode solutions to GitHub using **OAuth** and **GitHub Apps**.

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, SQL, Bash, HTML/CSS

Frameworks: React, Expo, Node.js, Next.js, Material-UI, Tailwind CSS, TensorFlow, Prisma ORM

Tools: Git, Docker, AWS, Vite, Redis, OAuth, MongoDB, MySQL, PostgreSQL, Retool, Postman